

**Solving Problems with Rational Numbers**

34 points

Name \_\_\_\_\_

Date \_\_\_\_\_

Period \_\_\_\_\_

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*Show your work. Circle your final answer.*

**This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.**

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**1. Operations in Scientific Notation**

12 points

1. Add or subtract. Write each answer in scientific notation.

a.  $(3.6 \times 10^5) + (2.1 \times 10^5)$  1 pt

b.  $(8.4 \times 10^{-6}) - (5.9 \times 10^{-6})$  1 pt

c.  $(7.5 \times 10^8) + (6.8 \times 10^8)$  1 pt

d.  $(4.2 \times 10^4) + (3.7 \times 10^3)$  1 pt

2. Multiply or divide. Write each answer in scientific notation.

a.  $(3 \times 10^4)(2 \times 10^6)$

1 pt

b.  $\frac{9.6 \times 10^9}{1.2 \times 10^4}$

1 pt

c.  $(5 \times 10^{-3})(4 \times 10^7)$

1 pt

d.  $\frac{1.8 \times 10^2}{6 \times 10^5}$

1 pt

3. Which is the value of  $(1.4 \times 10^3) + (2.9 \times 10^5)$  ?

1 pt

- ☐ A.  $3.04 \times 10^5$
- ☐ B.  $4.3 \times 10^5$
- ☐ C.  $2.914 \times 10^5$
- ☐ D.  $4.3 \times 10^8$

4. Select ALL of the expressions whose value is  $3 \times 10^6$  .

1 pt

- ☐ A.  $(3.2 \times 10^6) - (2 \times 10^5)$
- ☐ B.  $\frac{9 \times 10^9}{3 \times 10^4}$
- ☐ C.  $(6 \times 10^3)(5 \times 10^2)$
- ☐ D.  $(1.8 \times 10^6) + (1.2 \times 10^6)$
- ☐ E.  $(1.8 \times 10^6) + (1.2 \times 10^5)$
- ☐ F.  $\frac{9 \times 10^9}{3 \times 10^3}$

5. Kendra writes  $(2.3 \times 10^4) + (6.1 \times 10^6) = 8.4 \times 10^{10}$  . Name the two mistakes she made, and give the correct sum.

1 pt

6. A number is multiplied by  $4 \times 10^3$  and the product is  $2 \times 10^{-1}$  . What was the number?  
Write your answer in scientific notation.

1 pt

## 2. Real-World Problems and Significant Digits

9 points

7. Give the number of significant digits in each measurement.

a. 3,080 grams 1 pt

b. 0.00405 meters 1 pt

c. 12.500 seconds 1 pt

8. Solve. Report each answer with the correct number of significant digits.

a. A rectangular plate measures  $2.4 \times 10^2$  mm by  $1.35 \times 10^2$  mm. Find its area. 1 pt

b. A vehicle travels  $5.60 \times 10^2$  km in 8.0 hours. Find its average speed. 1 pt

9. Solve. Report each answer with the correct precision.

a. Two lengths are measured:  $6.25 \times 10^3$  mm and  $4.1 \times 10^2$  mm. Find their sum. 1 pt

b. Two masses are measured:  $9.30 \times 10^{-2}$  g and  $4.72 \times 10^{-3}$  g. Find their difference. 1 pt

10. Jonah adds two measured lengths, 4,560 m and 3.2 m, and says the sum must be written with two significant digits, as  $4.6 \times 10^3$  m. Explain why he is wrong, and give the correct sum. 1 pt

11. A rectangle's sides are measured as  $8.0 \times 10^1$  cm and  $3.15 \times 10^1$  cm. Which is the area, in scientific notation with the correct number of significant digits? 1 pt

- ☐ A.  $2.5 \times 10^3$  cm<sup>2</sup>
- ☐ B.  $2.52 \times 10^3$  cm<sup>2</sup>
- ☐ C.  $2.5 \times 10^2$  cm<sup>2</sup>
- ☐ D.  $25.2 \times 10^2$  cm<sup>2</sup>

### 3. The Order of Operations with Exponents and Radicals

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13 points

12. Evaluate each expression.

a.  $15 - 2 \cdot 3^2$  1 pt

b.  $\frac{5^2 + 7}{2^3}$  1 pt

c.  $36 \div 4 \cdot 3$  1 pt

d.  $-4^2 + (-4)^2$  1 pt

13. Evaluate each expression.

a.  $\sqrt{6^2 + 13}$  1 pt

b.  $\sqrt[3]{-125} + 2^{-2}$  1 pt

c.  $\left(-\frac{1}{3}\right)^2 + \sqrt{\frac{4}{9}}$  1 pt

14. A square patio of side 14 feet is sealed at 2.40 dollars a square foot, with a flat 85 dollars for delivery.

a. Write an expression for the total cost.

1 pt

b. Find the total cost.

1 pt

15. Which is the value of  $2 + 5 \cdot 3^2$  ?

1 pt

- ☐ A. 32
- ☐ B. 47
- ☐ C. 63
- ☐ D. 289

16. Select ALL of the expressions whose value is 6.

1 pt

- ☐ A.  $-\sqrt{36}$
- ☐ B.  $\sqrt{36}$
- ☐ C.  $36 \div 3 \cdot 2$
- ☐ D.  $\left(\frac{1}{6}\right)^{-1}$
- ☐ E.  $\sqrt{5^2 + 11}$
- ☐ F.  $\frac{4^2 + 8}{4}$

17. Nadia writes  $\sqrt{25 + 144} = \sqrt{25} + \sqrt{144} = 17$  . Explain why she is wrong, and give the correct value. 1 pt

18. A square tile has an area of 169 square inches. A second square tile has a side twice as long as the first. What is the area of the second tile? 1 pt